

FUNCTIONS OF MANGANESE (Mn)

It is a member of certain enzymes essential for protein synthesis.

Essential for certain nitrogen transformation in plants and micro-organisms.

DEFICIENCY OF MANGANESE

Pale-greenish-yellow discolorations between veins of young leaves.

FUNCTIONS OF COPPER (Cu)

It is a constituent of certain enzymes. Also necessary for photosynthesis.

It is involved in respiration and in the utilization of iron.

DEFICIENCY OF COPPER

Pale-green colour leaves.

The tips of older leaves dry off and die. Young leaves tips die-back.

FUNCTIONS OF ZINC (Zn)

Necessary for the action of certain enzymes. Essential for the formation of some growth hormones.

Important for the reproduction process of certain plants.

DEFICIENCY OF ZINC

production of mottled leaves. small leaves are also produced.

FUNCTIONS OF BORON(B)

It helps to promote the growth of tips of the roots.
The tips of shoots may also be promoted.

DEFICIENCY OF BORON:

Death of tips of roots and shoots.
Flower buds may fail to develop.
The rate of water absorption is reduced.
Translocation of sugars in plants is also reduced.
Poor growth and death of terminal buds in plants.

FUNCTIONS OF MOLYBDENUM (MOO₄)

Essential for nitrogen fixation in legumes.
Promotes nrate metabolism into amino-acids and proteins.
Increases the pH level in the soil.

DEFICIENCY OF MOLYBDENUM:

Retarded growth in legumes.
Necrosis of leaf tissue.

FUNCTIONS OF CHLORINE (Cl)

It influences photosynthesis and root growth.
It acts as counter ion for K and thereby contribute to osmotic potential.

DEFICIENCY OF CHLORINE:

Excessive branching and chlorosis of plants.
FUNCTIONS OF COBALT (Co).

DEFICIENCY:

It is essential for the symbiotic fixation of nitrogen in legumes.
lack of nitrogen fixation in legumes.
in legumes.

FACTORS AFFECTING NUTRIENTS -

AVAILABILITY IN THE SOIL:

Several factors are known to influence

the availability of nutrients to cultivated plants

these include:-

Soil pH:

The relative concentration of the hydrogen ions and or hydroxyl ions may either increase or decrease nutrients availability.

THE PRESENCE OF OTHER NUTRIENTS: The essence

of the various Minerals salts as analysed above under nutrients (Minerals salts) will either increase or decrease the effect Minerals salts in the soil.

LEACHING: is the washing down of the nutrients

in the top soil into the sub soil by water - thereby depriving plants of such nutrients. This may be in form of clean crop removal. Crops which lead to loss of nutrients in the soil.

Cleaning, deforestation or cutting down of harvested crops which lead to loss of nutrients in the soil.

Explain soil characteristics
influencing plant
growth and chemical
attributes.

ASSIGNMENT:

nutrients from one place to another, and
may occur under the influence of heavy rain
or wind storm.

EROSION: This helps in washing away of plant
nutrients from one place to another, and
may occur under the influence of heavy rain
or wind storm.

BURNING: This kills soil organisms, destroys the
texture and structure of the soil and finally
reduces plant nutrients.

OXIDATION AND REDUCTION: is the breakdown
of soil organic matter in the presence of oxygen.
Ammonium compounds in the soil may be
oxidised to amines, and nitrate may be
reduced to nitrogen or oxide of nitrogen.
(ie nitrous oxide) by denitrifying bacteria. The gases
produced from these reactions escape into the
atmosphere leaving the soil with a low
nitrogen content.

FERTILIZERS (Inorganic nutrients)

Fertilizers are chemical substances, generally in the form of powders, granules, pellets or crystals, which can be added to the soil when soil tests show that some important nutrients are lacking. They are usually solids, but may be dissolved and applied as liquids.

Types of Fertilizers:

There are two main types of fertilizers, they are simple and complex fertilizers.

Simple Fertilizers:

They contain or supply only one essential nutrient e.g.

- i. Ammonium nitrate — NH_4NO_3
- ii. Urea (from living things too) — $CO(NH_2)_2$ — 46% nitrogen.
- iii. Ammonium sulphate — $(NH_4)_2SO_4$ — 24% nitrogen.
- iv. Sodium nitrate — $NaNO_3$ — 16% nitrogen.
- v. Murate of potash — KCl — 50% potash.
- vi. Triple super phosphate — $(CaHPO_4)_3$ — 45% phosphate.
- vii. Single super phosphate — $CaHPO_4$ — 43.7% phosphate.

METHODS OF FERTILIZER APPLICATION

They are also known as compound or multiple fertilizers. They contain or supply two or more nutrients e.g. —

Broadcasting — This is suitable for fertilizers which are blown off by wind. The fertilizer is scattered randomly and fairly uniformly all over the ground by tossing. This can be done manually or mechanically by means of a machine called a fertilizer distributor. make sure the fertilizer does not have direct contact with the foliage because they are toxic to plants.

Ring method — This is the placement of fertilizers in a circular hole. The size of the hole depends on the type of roots of the plants —

i.	N.P.K — 10.10.10.
ii.	N.P.K — 15.15.15.
iii.	N.P.K — 5.10.5
iv.	N.P.K — 20.5.10
v.	N.P.K — 20.0.20
vi.	N.P.K.Mg — 10.15.10.2.

Row or side placement - This is the process of

making long lines along the farm land or ridge beside the cultivated crops and fertilizers later applied inside the line.

Band placement - is used when the plants are not planted in rows or are far away from one another at such a distance that the root hairs will be able to absorb it.

Top dressing - This is the supplementary application of fertilizers made to supplement the earlier ones.

Foliar spray - This is mostly used for leafy crops. The fertilizer is dissolved and later sprayed on the leaves of plants. This fertilizer is non-toxic to plants.

PRECAUTIONS TO BE TAKEN WHEN APPLYING FERTILIZERS

Fertilizers can only be applied when the farmland is weak and old.

When the soil is wet as a result of rain or irrigation, fertilizers can be applied.

Apply fertilizers about two weeks before planting or and some periods before fruiting or flowering.

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Before applying fertilizers, soil test must be

carried out to know the nutrient that is deficient. Always use ~~the~~ hand gloves and cover your nose when applying fertilizers because of their toxic nature.

CALCULATIONS OF FERTILIZERS

Complex

Formula for calculating fertilizers:

$\% N, P_2O_5, K_2O = \text{Fertilizer analysis}(\%) \times \text{wt. of part(g)}$

100kg

eg: calculate the % N, P_2O_5 and K_2O respectively in 25 kg bag each of urea (46% N), SSP (18% P_2O_5) and Mop (60% K_2O).

Solution:

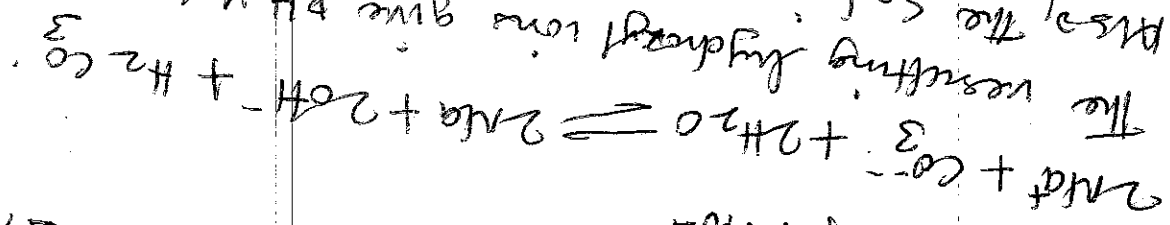
i. $\% N = \frac{46 \times 25}{100} = 11.5\%$

ii. $\% P_2O_5 = \frac{18 \times 25}{100} = 4.5\%$

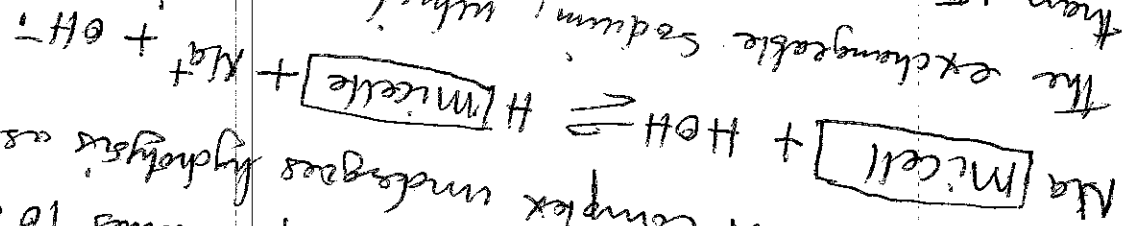
iii. $\% K_2O = \frac{60 \times 25}{100} = 15\%$

Sodic soils: These soils do not contain any great

amount of neutral soluble salts, the detrimental effects on plants being largely due to the toxicity of the sodium as well as of the hydroxyl ions. The high pH is large due to the hydrolysis of sodium carbonate, which occurs as follows:-



The resulting hydroxyl ions give pH values 10 and above. Also, the sodium complex undergoes hydrolysis as follows:-



The exchangeable sodium, which occupies decidedly more than 15 percent of the total exchange capacity of these soils is free to hydrolyse because the concentration of neutral soluble salts is rather low. The electrical conductivity of a saturated extract is above 8.5 often rising as high as 10.0. Such soils usually are in an unsatisfactory physical condition.

As already stated, the leaching of a saline-sodic soil will readily change it to a characteristic -

Sodic soil.

Because of the extreme alkalinity resulting from the Na_2CO_3 present, the surface of soils -

carried upward by the capillary water - hence the name black alkali is frequently used. These soils are

often located in small areas called slick spots -

surrounded by soils that are relatively productive. (41)

SALINE SOILS: These soils contain a concentration of neutral soluble salts sufficient to seriously interfere with the growth of most plants. The electrical conductivity of a saturated extract (ECe) is greater than 4 mhos/cm. Less than 15 percent of the cation exchange capacity of these soils is occupied by sodium ions and the pH usually is below 8.5. This is because the soluble salts present are mostly neutral, and because of their domination, only a small amount of exchangeable sodium is present. Such soils are sometimes called white alkali soils because a surface incrustation, if present, is light in colour. The excess soluble salts, which are mostly chlorides and sulfates of sodium, calcium and magnesium, can readily be leached out of these soils with no appreciable rise in pH. This is a very important practical consideration in the management of these soils. Care must be taken to be certain that the leaching water is low in sodium.